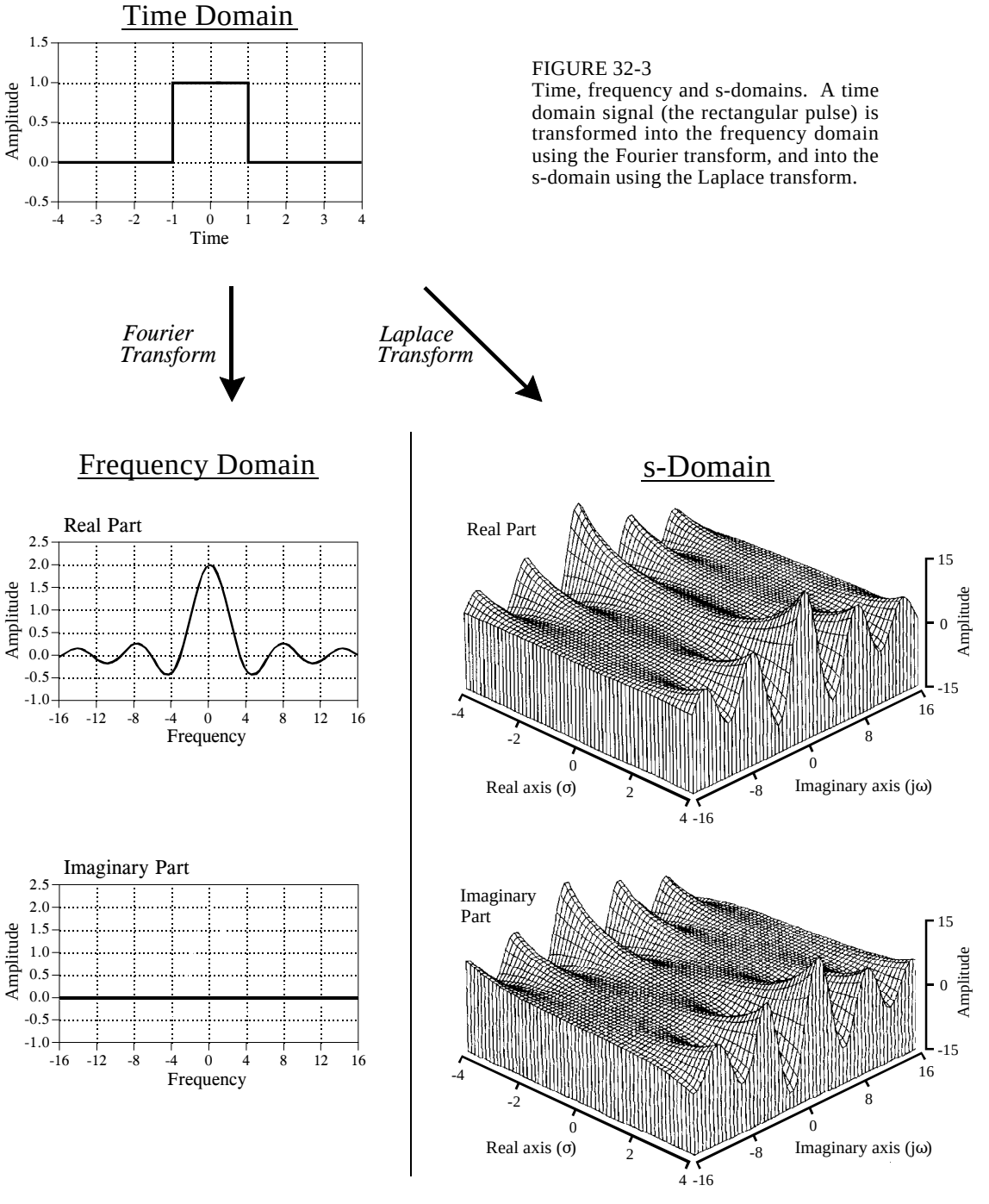


FIGURE 32-3
Time, frequency and s-domains. A time domain signal (the rectangular pulse) is transformed into the frequency domain using the Fourier transform, and into the s-domain using the Laplace transform.

The topographical surfaces in Fig. 32-3 are graphs of these equations. These equations are quite long and the mathematics to derive them is very tedious. This brings up a practical issue: with algebra of this complexity, how do we know that we haven't made an error in the calculations? One check is to verify